TERM PROJECT

"Does Higher Internet Penetration Cause Lower Unemployment?"

Data Analysis 4

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Vienna, Austria

April 2025

Introduction

The spread of the internet in the last 20 years has changed recruiting practices and methods used by individuals searching for employment opportunities. Internet availability can help decrease frictions and transaction costs in job search and matching process, thus helping to reduce unemployment. However, the identification of the causal impact of the internet is complicated since richer and economically developed countries have both quicker access to the internet and lower unemployment rates independently of internet presence.

Does internet access cause unemployment rates to fall? The research question is very relevant to practice since governments allocate large sums to the development of internet access in an effort to stimulate positive labor market outcomes. Whether there is a causal link between the variables is important to verify because it can validate such policies.

In order to investigate the relation between the internet penetration and unemployment rates, a panel database on country-level indicators published by the World Bank was used for the time frame 2000-2022 and includes 219 countries (4,718 observations in total). Internet penetration rates are considered the causal variable (X), while the unemployment rate is considered the variable of interest (Y). In order to control for time invariant unobserved heterogeneity among countries and time trends, a two-way fixed-effects model was estimated with fixed effects at the country and year levels.

The key result is that, after controlling for fixed effects, internet penetration does not affect unemployment significantly (coefficient = -0.0016, se = 0.0080, p = 0.845). However, the non-significant impact turns out to be heterogeneous depending on income level of the country. The coefficient is nearly four times bigger among rich countries (coefficient = -0.039) than in poorer economies (coefficient = -0.010).

The remainder of the paper is organized as follows. Section 2 describes the data used in the study. Section 3 explains the identification strategy. Section 4 presents empirical findings and provides discussion. Section 5 includes robustness checks and heterogeneity analyses.

Data

2.1 Sources

Both variables are taken from the open data platform of the World Bank, which makes it possible to use international comparable data for both the dependent and explanatory variables. Y is unemployment in percent of the total labor force, measured according to International Labour Organization (ILO) estimates (Indicator SL.UEM.TOTL.ZS). X is internet users as a percent of the total population (Indicator IT.NET.USER.ZS). The indicators can be downloaded free-of-charge without any log-in from data.worldbank.org.

2.2 Sample

The database includes 17,556 observations across both indicators. After limiting the time span from 2000 to 2022 and excluding cases in which there were missing values on one of the indicators (internet usage: 1,000 missing cases, unemployment rate: 714 missing), 4,718 observations remained for analysis across 219 countries, creating an unbalanced panel. The data missing appears primarily among small countries with low capacity for statistics and the early years when internet use measurements were scarce.

2.3 Descriptive Statistics

Table 1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max	Obs.
Unemployment rate (%)	7.83	5.70	0.10	37.32	4,718
Internet users (% pop)	35.78	30.58	0.00	100.00	4,718

Source: World Bank Open Data.

The mean level of unemployment for the sample group stands at 7.83% (standard deviation = 5.70), varying from 0.10% to 37.32%. The average internet usage rate comes out at 35.78% (standard deviation = 30.58%). The substantial standard deviation stems from the difference between the early adopters who belong to developed nations and the late adopters who hail from developing nations. The significant within-variation of the data is exactly what allows for identification in the panel fixed effects setting.

Figure 1 (Panel A) highlights a near-flat OLS line fit in the cross-sectional scatter plot, indicating the lack of any unconditional relationship between internet usage and unemployment rates. Panel B captures the trends on a global scale; both internet usage rates and unemployment rates show a consistent upward trend, with internet usage increasing from around 8% in the year 2000 to 65% by 2022. In contrast, unemployment levels experienced cyclical trends, with the highest peaks observed during the 2008 financial crisis and the 2020 pandemic.

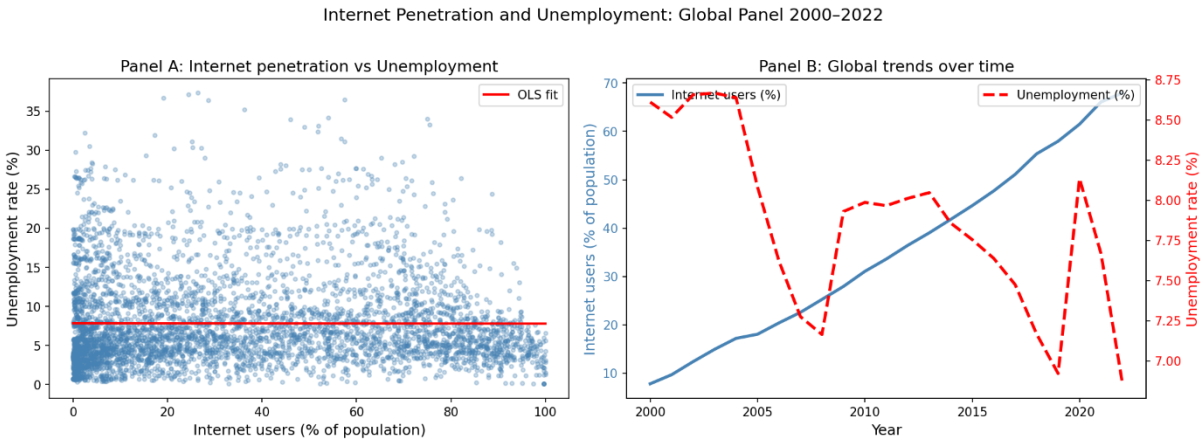

Figure 1: Internet penetration and unemployment.
Panel A: raw scatter with OLS fit. Panel B: global average trends over time.
Source: World Bank.

3. Identification Strategy

Endogeneity is the primary issue with regards to establishing a relationship between internet penetration rate and unemployment rates since economically strong nations are more likely to have better internet access while concurrently experiencing low levels of unemployment due to other external factors. There can be reverse causality as well where the government makes investments in digitization at times of economic crises.

For dealing with these issues, the model used primarily is TWFE panel regression:

$$\text{Unemployment_ct} = \alpha + \beta \cdot \text{Internet_ct} + \delta_c + \lambda_t + \varepsilon_ct$$

Where c is the country subscript and t is the year subscript. The country fixed effects δ_c control for all time-invariant variables, such as institutional quality, geographic characteristics, culture, and the level of development in the past, which may bias the results. Year fixed effects λ_t control for all shocks that affect all countries simultaneously, such as the financial crisis in 2008 and the recent coronavirus outbreak. Thus, the β parameter estimates the causal effect of internet usage growth on unemployment within a country, net of these confounders.

The identification strategy rests on the assumption that within a country, apart from country and year fixed effects, the growth in the internet population is orthogonal to other variables which determine the unemployment rate. It can be assumed since internet infrastructure is developed in advance and not in response to any other economic variables. Serial correlation within each country is controlled by clustering standard errors at the country level.

4. Results

4.1 Main Results

Table 2 shows the key findings in four models. The first model (Model 1) is a basic OLS regression without any controls. The second model (Model 2) incorporates year fixed effects to address any trends that might affect countries globally throughout the period under investigation. Model 3 is the best model since it includes country fixed effects to address all unchanging factors within each country.

Table 2: Main Results

	Model 1 OLS	**Model 2 OLS + Year FE**	**Model 3 Panel FE**	**Model 4 Panel FE (log)**
Internet (%)	-0.0005	0.0090***	-0.0016	
(se)	(0.0027)	(0.0033)	(0.0080)	
Log Internet				0.2632
(se)				(0.1945)
Country FE	No	No	Yes	Yes
Year FE	No	Yes	Yes	Yes
Observations	4,718	4,718	4,718	4,718
R-squared	0.000	0.010	0.000	0.003

*Note: Dependent variable is unemployment rate (%). Standard errors clustered at country level in Models 3–4. * p<0.1, ** p<0.05, *** p<0.01.*

The best-fitting model (Model 3) shows an insignificant coefficient of −0.0016 for internet penetration (se = 0.0080, p = 0.845). In other words, controlling for all time-invariant country features and global time trends, internet penetration has no statistically significant effect on the level of unemployment.

Going through the models provides some interesting conclusions. First, when considering the basic OLS estimation model (Model 1), the coefficient will be near zero (−0.0005), which can be seen graphically from the flat OLS curve in Figure 1. The inclusion of year fixed effects (Model 2) leads to a positive and statistically significant coefficient (0.0090***). This happens since the correlation between internet penetration and unemployment is positive for each year, implying that nations having more internet penetration also have higher unemployment levels, most likely due to the fact that high-income nations provide more accurate measures of their

unemployment. When using country fixed effects, we observe differences rather than the actual within-country causal impact. The log transformation (Model 4) results in a positive but insignificant coefficient of 0.2632 (se = 0.1945).

4.2 Discussion

Is this null result in Model 3 due to no causality, or is it due to low statistical power? The 95% confidence interval for Model 3 is between −0.0172 and 0.0141, so we can reject anything above about 1.7 percentage points per 10 percentage points of increased internet usage rate – which is quite a tight estimate. Thus, it seems that this result is an indication of no causality of any significant size.

As compared to Atasoy (2013), who found an increase in the employment rate by 1.8 percentage points due to broadband infrastructure growth in the US counties, the null finding could be attributed to several factors. First of all, the latter study utilizes county-level data, whereas we use countries, and this might mean that effects can cancel each other out in our case. Also, a high portion of developing countries in our sample, where there might be certain obstacles on the way to formal employment, could play a role here.

5. Robustness, Heterogeneity, and Policy Conclusions

5.1 Robustness Checks

The following two robustness tests verify the reliability of the primary null hypothesis finding. First, the estimation using data on pre-COVID years only (2000-2019) provides a coefficient estimate of −0.0040 (se = 0.0090, N = 4,106). In this case, the coefficient value is somewhat more negative but insignificant, thereby ruling out the influence of the shock caused by the outbreak of the virus on both IT adoption patterns and labor market dynamics.

Second, the test uses a more balanced dataset obtained after omitting countries with less than the lowest 10% number of observations, resulting in the coefficient estimate of −0.0020 (se = 0.0080, N = 4,651), which is very close to the previous one. As seen from the results of the robustness tests, the coefficient stays negative, small, and insignificant.

5.2 Heterogeneity

Although the overall effect size is statistically zero, there is some theoretically interesting heterogeneity found. When divided by income, the effect of internet penetration on unemployment rate within each country is −0.0390 (standard error=0.0439, n=825) for high-income nations and −0.0103 (standard error=0.0085, n=3,893) for low and middle-income nations. Although none are statistically significant, the effect size from the latter category is almost four times smaller than the former category.

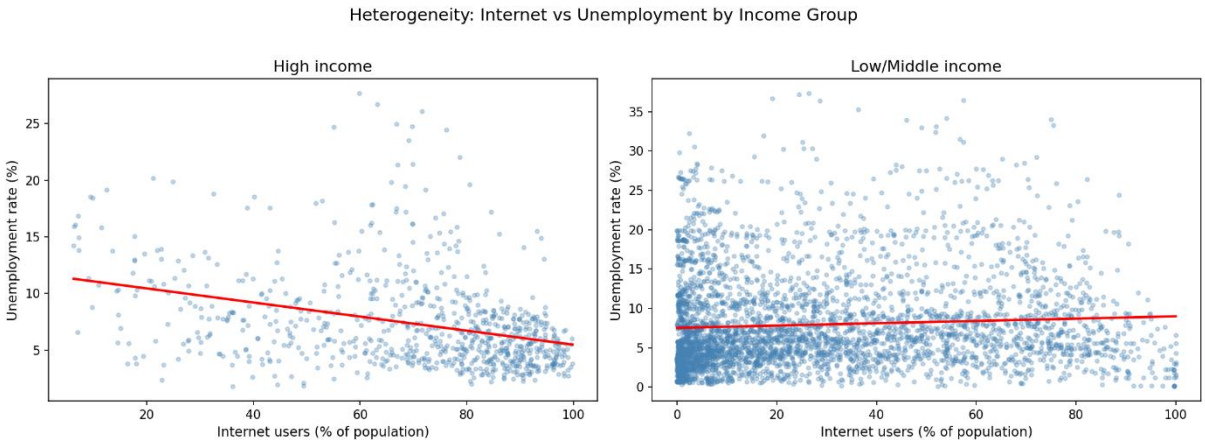

Figure 2: Heterogeneity by income group. Left panel: high-income countries show a clear downward trend. Right panel: low/middle-income countries show a near-flat relationship. Source: World Bank.

The relationship described above is illustrated by Figure 2. On the left-hand side (high income countries), the unadjusted scatter plot indicates a very clear negative slope, where countries with more internet users tend to be associated with fewer unemployed individuals. The regression line in the graph on the right-hand side (low- and middle-income countries), on the other hand, almost parallels the x-axis. This finding supports the complementarity theory in that internet use lowers unemployment only if labor market structures are sufficient.

5.3 Policy Conclusions

The findings of this paper indicate that the assertion that the internet is a powerful universal tool to reduce unemployment does not hold any weight. First, it turns out that internet penetration as a cause is not strongly correlated with unemployment levels when country-specific factors are controlled. Thus, billions of dollars spent on broadband policies worldwide may simply be wasted.

At the same time, there are certain nuances regarding the causal relationship between the two phenomena. Internet investments in high-income countries with efficient labor markets seem to have a negative impact on unemployment even though it is hard to detect the exact magnitude of this relationship. For middle/lower income countries, the investment in internet alone cannot result in unemployment decline without any other improvements (training, education, etc.).

To better determine causality, one can rely on regional or individual-level data and use a natural experiment, for instance, a staggered rollout of fiber networks or subsidies in order to estimate the causal effect. This research would resemble the design employed by Atasoy (2013).

References

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